

Array Questions Practice Sheet (Easy to Medium)

These questions focus mainly on basic array logic without advanced techniques like dynamic programming, graphs, tries, segment trees, heaps, or complex algorithms. Great for strengthening loops, conditions, indexing, brute force thinking, and basic problem-solving.

1. Basic Traversal & Operations

- 1 Find the maximum element in an array
- 2 Find the minimum element in an array
- 3 Calculate the sum of all elements
- 4 Calculate the average of array elements
- 5 Count even and odd numbers
- 6 Search an element using linear search
- 7 Find the index of a given element
- 8 Print array elements in reverse order
- 9 Copy one array into another
- 10 Count positive, negative, and zero elements

2. Frequency & Counting Problems

- 1 Count frequency of each element
- 2 Find duplicate elements in an array
- 3 Find the first repeating element
- 4 Find the first non-repeating element
- 5 Count occurrences of a given number
- 6 Find the most frequent element
- 7 Find the least frequent element
- 8 Remove duplicate elements
- 9 Check whether all elements are unique
- 10 Count pairs with a given sum (brute force)

3. Array Rearrangement

- 1 Reverse an array
- 2 Rotate array left by one position
- 3 Rotate array right by one position
- 4 Move all zeros to the end

- 5 Move all negative numbers to one side
- 6 Swap alternate elements
- 7 Separate even and odd elements
- 8 Rearrange array in ascending order
- 9 Rearrange array in descending order
- 10 Shift all elements by K positions

4. Subarray & Window Basics

- 1 Find sum of all subarrays
- 2 Find the largest sum subarray using brute force
- 3 Find the smallest sum subarray
- 4 Print all subarrays
- 5 Find subarray with a given sum using loops
- 6 Count total number of subarrays
- 7 Find longest subarray of positive numbers
- 8 Find shortest subarray with sum greater than X
- 9 Find maximum product subarray using brute force
- 10 Print all subarrays of size K

5. Sorting Based Questions

- 1 Implement bubble sort
- 2 Implement selection sort
- 3 Implement insertion sort
- 4 Check if array is sorted
- 5 Merge two arrays
- 6 Find the second largest element
- 7 Find the second smallest element
- 8 Find the third largest element
- 9 Sort only even numbers in array
- 10 Sort only odd numbers in array

6. Prefix/Suffix Basic Thinking

- 1 Find prefix sum array
- 2 Find suffix sum array
- 3 Find equilibrium index

- 4 Find product of all elements except self using loops
- 5 Find left maximum for every element
- 6 Find right minimum for every element
- 7 Find running sum of array
- 8 Replace every element with greatest element on right side
- 9 Find difference between prefix and suffix sums
- 10 Check if array can be split into two equal sum parts

7. 2D Array Questions

- 1 Print matrix row-wise
- 2 Print matrix column-wise
- 3 Find sum of all matrix elements
- 4 Find transpose of matrix
- 5 Print diagonal elements
- 6 Find largest element in matrix
- 7 Find smallest element in matrix
- 8 Add two matrices
- 9 Multiply two matrices
- 10 Search element in 2D array

8. Pattern & Logic Building

- 1 Find missing number from 1 to N
- 2 Find all pairs in array
- 3 Find all triplets in array
- 4 Check if array is palindrome
- 5 Find intersection of two arrays
- 6 Find union of two arrays
- 7 Find common elements in three arrays
- 8 Check if two arrays are equal
- 9 Find majority element using brute force
- 10 Find leaders in array

Suggested learning order:

1. Traversal Questions
2. Frequency Questions
3. Rearrangement Questions
4. Sorting Questions
5. Subarray Questions
6. Prefix/Suffix Thinking

7. 2D Arrays

8. Mixed Logic Problems